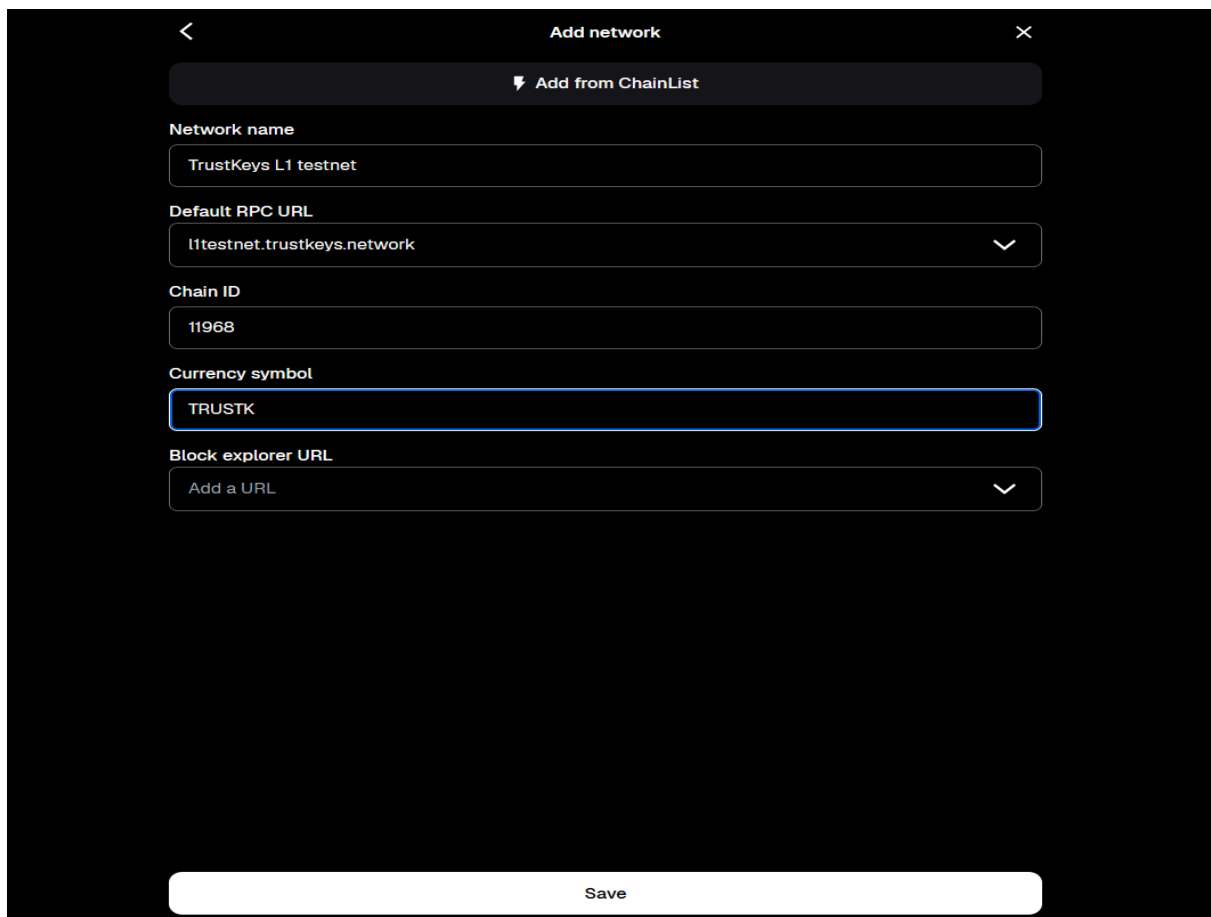
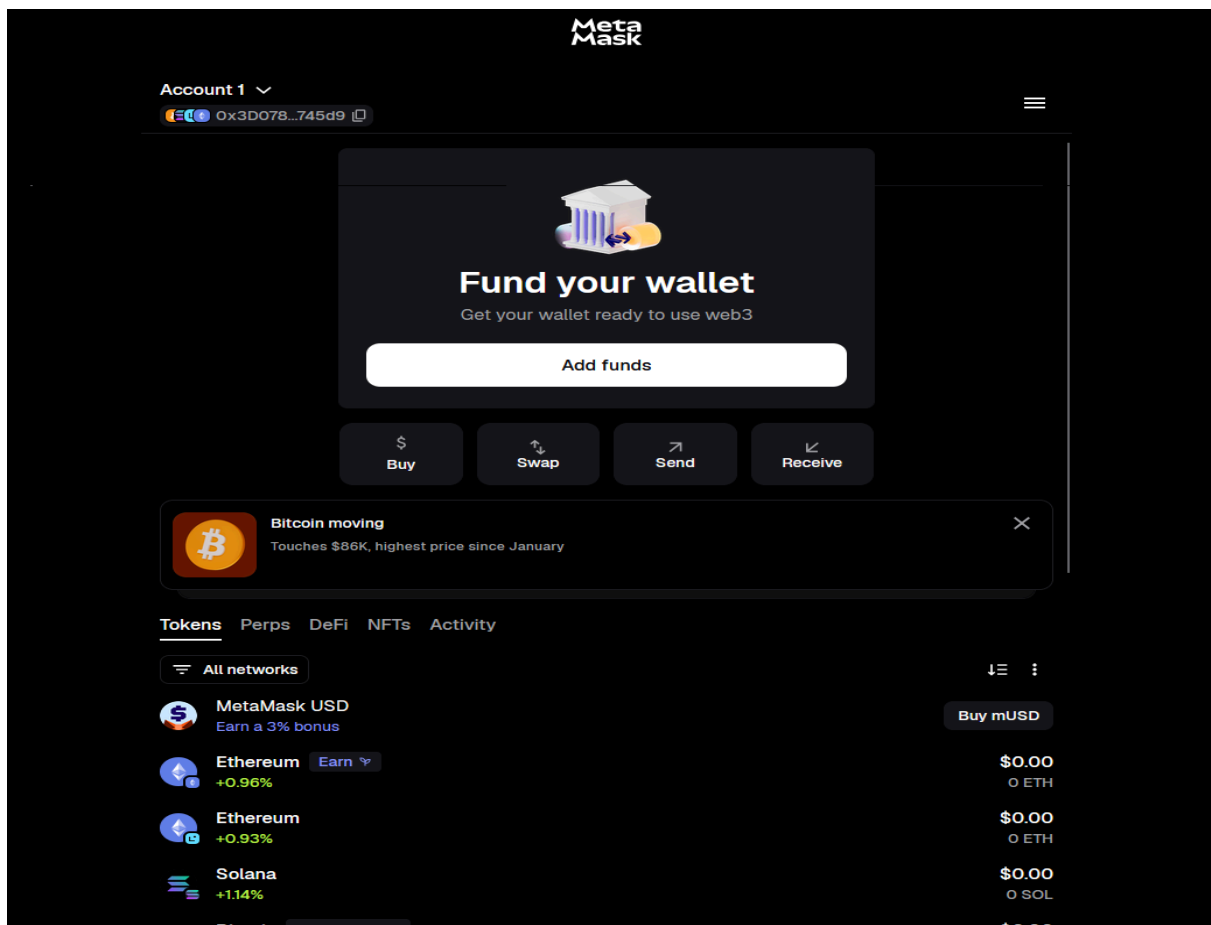
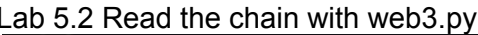


## Lab 5.1 Add Trustkeys L1 to MetaMask





The screenshot shows a Visual Studio Code editor window with a file named `fee_history.py` open. The code is written in Python and interacts with a blockchain network via Web3.js. It prints details about the latest block and iterates through the last 20 blocks to print their base fees and gas used ratios.

```
D:\> SCHOOLWORK > Blockchain > fee_history.py  
17  
18     print("Latest block:", blk["number"])  
19     print("Gas limit:", blk["gasLimit"])  
20     print("Gas used:", blk["gasUsed"])  
21     print("Base fee:", blk["baseFeePerGas"], "wei")  
22  
23     fh = w3.eth.fee_history(  
24         20,  
25         "latest",  
26         [10, 50, 90]  
27     )  
28  
29     print("\nLast 20 blocks:")  
30  
31     for i, base in enumerate(fh["baseFeePerGas"][::-1]):  
32         block_number = fh["oldestBlock"] + i  
33         gas_ratio = fh["gasUsedRatio"][i]  
34  
35         print(  
36             block_number,  
37             "baseFee =", base,  
38             "gasUsedRatio =", gas_ratio  
39         )
```

The terminal at the bottom shows the execution results:

```
PS D:\SCHOOLWORK> cd Blockchain  
PS D:\SCHOOLWORK\Blockchain> python fee_history.py  
Connected: True  
Chain ID: 11968  
Latest block: 717544  
Gas limit: 30000000  
Gas used: 24587  
Base fee: 8 wei  
  
Last 20 blocks:  
717525 baseFee = 8 gasUsedRatio = 0.1  
717526 baseFee = 8 gasUsedRatio = 0.0038004666666666665  
717527 baseFee = 8 gasUsedRatio = 0.0199888  
717528 baseFee = 8 gasUsedRatio = 0.0007  
717529 baseFee = 8 gasUsedRatio = 0.0007  
717530 baseFee = 8 gasUsedRatio = 0.0040437666666666667  
717531 baseFee = 8 gasUsedRatio = 0.0015531666666666667  
717532 baseFee = 8 gasUsedRatio = 0.16666666666666666  
717533 baseFee = 8 gasUsedRatio = 0.16666666666666666  
717534 baseFee = 8 gasUsedRatio = 0.16666666666666666  
717535 baseFee = 8 gasUsedRatio = 0.16666666666666666  
717536 baseFee = 8 gasUsedRatio = 0.16666666666666666  
717537 baseFee = 8 gasUsedRatio = 0.16666666666666666  
717538 baseFee = 8 gasUsedRatio = 0.16666666666666666
```

Problems 2 Output Terminal ...

 powershell     ... | 

```
PS D:\SCHOOLWORK\Blockchain>
```

Lab 5.3: Sending 0.001 To Bui Quang Minh

Sending

0.001 TRUSTK

From

Account 1

To

Ox9f6ed78099f6C6822D9b885770cDd3239DF5DA56

Network

T TrustKeys L1 testnet

Network fee ?

< 0.0001 T TRUSTK

Speed

|                |                                       |
|----------------|---------------------------------------|
| ?              | -0.001 TRUSTK                         |
| Status         | Confirmed                             |
| Date           | Sep 23, 2026, 1:30 PM                 |
| From           | <div></div> namtran (Ox3d078...745d9) |
| To             | <div></div> Ox9f6ed...5da56           |
| Network        | T TrustKeys L1 testnet                |
| Transaction ID | Ox188f9...02284                       |
| Total amount   | 0.001 TRUSTK                          |

```

56 blk = w3.eth.get_block(rcpt["blockNumber"])
57
58 base_fee = blk["baseFeePerGas"]
59 gas_used = rcpt["gasUsed"]
60 eff_price = rcpt["effectiveGasPrice"]
61
62 paid = gas_used * eff_price
63 burned = gas_used * base_fee
64 tip = gas_used * (eff_price - base_fee)
65
66 print("\n--- Fee decomposition ---")
67 print("Transaction type:", tx["type"])
68 print("Block number:", rcpt["blockNumber"])
69 print("Gas used:", gas_used)
70 print("Base fee:", base_fee, "wei")
71 print("Effective gas price:", eff_price, "wei")
72
73 print("Paid:", paid, "wei")
74 print("Burned:", burned, "wei")
75 print("Tip:", tip, "wei")
76
77 print("CHECK:", paid == burned + tip)

```

Problems 2 Output Terminal ... powershell ⚠ + ▾ 🗑 ...

```

PS D:\SCHOOLWORK\Blockchain> python fee_history.py
--- Fee decomposition ---
Transaction type: 2
Block number: 717545
Gas used: 21000
Base fee: 8 wei
Effective gas price: 2000000000 wei
Paid: 420000000000000 wei
Burned: 168000 wei
Tip: 41999999832000 wei
CHECK: True
PS D:\SCHOOLWORK\Blockchain>

```

Decompose the fee

## Lab 5.4: Trace a contract interaction on Sepolia Etherscan

② Transaction Fee:

0.00006830468401005 ETH

② Gas Price:

1.09541631 Gwei (0.00000000109541631 ETH)

② Gas Limit & Usage by Txn:

228,782 | 62,355 (27.26%)

② **Gas Fees:**

Base: 0.97541631 Gwei | Max: 1.291391094 Gwei | Max Priority: 0.12 Gwei

② **Burnt & Txn Savings Fees:**

🔥 Burnt: 0.00006082208401005 ETH (\$0.00)

Txn Savings: 0.00001222000765632 ETH (\$0.00)

② Other Attributes:

Txn Type: 2 (EIP-1559)

Nonce: 7659

Position In Block: 116

② Input Data:

| # | Name        | Type    | Data                                                               |
|---|-------------|---------|--------------------------------------------------------------------|
| 0 | _to         | bytes32 | 0x2dc564f07856a5418cae8a51d3d1e4faa4951dccc98659c6b1c15adbfc91d5dc |
| 1 | _amount     | uint256 | 100000000000000000000                                              |
| 2 | _secrethash | bytes32 | 0x08639575532fb17a757b8a780ad3ff1126c9f006bb09686f0e760e32e5e5817b |

 Switch Back

 Export JSON

[View In Decoder](#)

### More Details:

— Click to show less

🔗 A transaction is a cryptographically signed instruction that changes the blockchain state. Block explorers track the details of all transactions in the network. [Learn more about transactions in our Knowledge Base.](#)

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Got it!

② Other Attributes:

Txn Type: 2 (EIP-1559)

Nonce: 7659

Position In Block: 116

② Input Data:

```
Function: depositToAztecPublic(bytes32 _to,uint256 _amount,bytes32
secretHash)
```

MethodID: 0x284b5dc6

```
[0]: 2dc564f07856a5418cae8a51d3d1e4faa4951dccc98659c6b1c15adbfc91d5dc
```

[illegible]View Input As  Decode Input Data

 Export JSON

[View In Decoder](#)

 Write Contract

### More Details:

— Click to show less

## Transaction Receipt Event Logs

164

### Address

0xa8c3a167638ac2c3d4d52be193cfd1bc6e5af66    

### Name

Transfer (index\_topic\_1 address from, index\_topic\_2 address to, uint256 value) [View Source](#)

### Topics

0 0xddf252ad1be2c89b69c2b068fc378daa952ba7f163c4a11628f55a4df523b3ef

1: from Dec  0x9d748e0BD6af022DeE9F2E88125310204Ba7437F

2: to Dec  0x5bc4B38bC617523b4EaecFEa3318d0E57728B5DE

### Data

value (uint256) : 100000000000000000000

Dec Hex

165

### Address

0x88757f6c7d63e2d9c661674b5e799e9ef3ce9395    

### Topics

0 0x6406de2408cbb641d69139ebf977ffd682d709ede2d7b1009509ca5e7426a02e

1  0x00e0b4c3ff750f797ec01e6a4bb7536052d2ae8fe1ff8ae581092c31d2f08c36

### Data

Hex  002b968dbb0a0566048630ec56d154dc3cb6aa0 Dec Hex  
4253ee9a84137576fa61b13d0  
Hex  0000000000000000000000000000000000000000000000000000000000000000  
0000000002  
Hex  0000000000000000000000000000000000000000000000000000000000000000  
0000000003  
Hex  0000000000000000000000000000000000000000000000000000000000000000  
0000aa36a7  
Hex  0000000000000000000000000000000000000000000000000000000000000000  
0000000003

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Got it!

### Decoded Results:

Function `depositToAztecPublic`

bytes32 \_to

0x2dc564f07856a5418cae8a51d3d1e4faa4951dccc98659c6b1c15adbfc91d5dc

uint256 \_amount

1000000000000000000000000

Wei

bytes32 \_secretHash

0x08639575532fb17a757b8a780ad3ff1126c9f006bb09686f0e760e32e5e5817b

Copy All Parameters